Vladislav Korzh

Senior AI Engineer

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Professional Profile

Senior AI Engineer with 7+ years of experience delivering NLP, information retrieval, document intelligence, enterprise RAG, and controlled agentic workflows across telecom, automotive engineering, financial document operations, and enterprise information management. Strongest where model behavior meets software engineering: benchmark construction, retrieval and reranking, confidence policies, human review, typed tool contracts, and trace-level evaluation — treating abstention, auditability, and deterministic fallback as core product capabilities.

Experience

M-Files Full-time | Remote (Finland)

Senior AI Engineer / AI Platform Lead *Apr 2024 – Present*

- Designed permission-aware hybrid retrieval combining BM25, dense vectors, structured filters, rank fusion, and cross-encoder reranking while preserving tenant and ACL constraints across every stage.
- Defined versioned chunk, citation, and evidence contracts; pilots held citation coverage near 85–90%, and source-backed research fell from tens of minutes to 3–5 minutes.
- Extended the platform with a bounded LangGraph supervisor and three read-only specialists for retrieval, comparison, and evidence verification, using typed artifacts instead of free-form agent history.
- Built golden and adversarial evaluations for routing, groundedness, refusal, tool arguments, and recovery; shipped behind feature flags with canary and rollback gates and search-only fallback.

Rossum Full-time | Remote (Czech Republic)

Applied ML Engineer → Senior Applied ML Engineer *Jul 2022 – Mar 2024*

- Built training and evaluation pipelines for multilingual document routing and layout-aware extraction using OCR tokens, bounding boxes, transformer encoders, and field-level schemas.
- Introduced supplier-grouped and temporal splits to prevent template leakage, plus slice reports for rare layouts and OCR failure modes.
- Implemented calibrated field confidence, deterministic validation of totals, dates, and currencies, and routing of uncertain documents into human review queues.
- Captured operator corrections as versioned feedback, grew the golden set from real exceptions, and used shadow rollout before changing production thresholds.

Vector Informatik Full-time | Remote (Germany)

Search / NLP Engineer *Jan 2021 – Jun 2022*

- Built engineering knowledge search over GitLab repositories and docs with incremental, ACL-aware ingestion and continuation after individual repository failures.
- Moved from a BM25 baseline to hybrid retrieval with Sentence-BERT and rank fusion: Recall@20 grew from about 0.71 to 0.82, and to 0.86 with top-N cross-encoder reranking.
- Added BM25 fallback and dashboards for MRR@10, nDCG@10, zero-result rate, index freshness, and P95 latency; automated audit snapshots from hours to minutes.

Comarch Full-time | Remote (Poland)

Junior Backend / ML Engineer → ML Engineer *Feb 2019 – Dec 2020*

- Built failure-intelligence tooling for telecom billing regressions: log normalization plus TF-IDF word/char n-gram classifiers with class-weighted logistic regression.
- Designed build-aware temporal splits and evaluated macro-F1, PR-AUC, and coverage at confidence thresholds; versioned data, thresholds, and models in MLflow.
- Shipped an inference API with similar-case retrieval feeding Allure reports, reducing initial triage time by an estimated 25–35% while retaining manual review.

Skills

Languages: Python, SQL, Bash
ML / NLP: PyTorch, Hugging Face Transformers, SentenceTransformers, LayoutLM, scikit-learn, pandas, NumPy, MLflow
Search / RAG: OpenSearch, Elasticsearch, Qdrant, BM25, Dense Retrieval, Cross-Encoder Reranking, RRF, LangGraph
Backend / Data: FastAPI, Pydantic, PostgreSQL, Redis, Kafka, S3, REST APIs, SSE, JSON Schema
MLOps / Observability: Docker, Kubernetes, GitHub Actions, GitLab CI, Langfuse, OpenTelemetry, Prometheus, Grafana
Evaluation: Golden & Adversarial Sets, Recall@k, MRR, nDCG, Calibration, Shadow & Canary Rollouts

Education

ITMO University

Master's Degree, Computer Science, GPA: 5.0/5.0 *Sep 2024 – Jul 2026*